
PRODUCT UX AUDIT

LexAmplify

Software Platform

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Platform **React 18 + Flask (Python)**

1. Executive Summary

LexAmplify is a full-stack AI-powered legal practice management platform purpose-built for Indian advocates and law firms. It integrates an AI command palette, contract analysis, conflict checking, a document vault, a legal calendar, court resource browsing, and a virtual courtroom simulation into a single unified enterprise console.

This report audits every module that has been built, catalogues the exact features delivered, identifies the remaining gaps, and provides a prioritised build list for the next phase of development.

2. Technical Architecture

2.1 Frontend Stack

Component	Detail
Framework	React 18 + Vite (SPA)
Routing	React Router v6 with nested layouts
Theming	CSS custom properties + data-theme attribute on <html>
State Management	useState / useReducer per component — no Redux
AI Streaming	Server-Sent Events (SSE) via ReadableStream
Icons	Inline SVG — zero external icon dependency
Fonts	Georgia (serif headings) + System-UI (body)
Deploy Target	Firebase Hosting (lexamplify.web.app)

2.2 Backend Stack

Component	Detail
Framework	Python Flask — single create_app() factory in app.py
Database	SQLite (lex_assistant.db) — adjacency-list folder tree
AI / LLM	Groq (Llama / Mixtral) for all generation tasks
Legal Search	Tavily + Indian Kanoon API for live precedent lookup
Authentication	Flask-JWT-Extended with Bearer token scheme
PDF Generation	ReportLab (pure Python — no wkhtmltopdf binary)
DOCX Generation	python-docx (pure Python — no Pandoc binary)

Deploy Target	Render.com (lexamplify-backend.onrender.com)
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3. Module Status

The following table covers every user-facing module, its route, what it delivers, and its current completion status.

Module	Description	Status
Landing Page <i>/ (public)</i>	Dark glassmorphic hero with split layout, animated terminal mockup, feature showcase, testimonials, pricing teaser, sticky header, light/dark theme toggle, and CTA buttons to Login.	Complete
Authentication <i>/login /signup</i>	Split-panel login with branded left panel (hidden on mobile), JWT tokens stored in localStorage, themed form fields, error messaging. Sign-up exists. No password reset or email verification yet.	Partial
Dashboard <i>/dashboard</i>	Live triage stats (limitation expiries, pending judgments, draft counts, tracked cases) from API. CNR number sync bar linking to eCourts. Morning brief of 48-hour events. Quick-action tiles for all tools. Zero hardcoded data.	Complete
Universal Agent <i>⌘K global overlay</i>	Full-screen AI command palette with SSE streaming. Multi-session sidebar (pin, rename, share, delete). Rich-text draft editor with toolbar. File upload mid-conversation. Save-to-Vault with PDF/DOCX/Native format selector. AI provenance audit trail. LLM intent-driven navigation to any module.	Complete
Contract Analyzer <i>/contract-analyzer</i>	Upload PDF/DOCX → AI risk analysis with colour-coded risk pills (High/Amber/Green). Collapsible summary banner. Clause-level risk annotation sidebar. AI-powered clause rewrite. One-click export. Summary and Recommendations tabs. Chat-with-contract panel.	Complete
Court Resources <i>/court-resources</i>	Tabbed browser: live court session indicator (In/Holiday), district court data, IPC/BNS/CrPC/PoSH/FEMA statute lookup, court fee calculator, IP asset tracker (full CRUD), cause-list pull from eCourts API.	Partial
Conflict Engine <i>/conflict-engine</i>	Name-check tab and multi-document cross-check tab. Batch file upload (multiple PDFs/DOCX). Deduplication with yellow warning toast. Drag-and-drop slots with progressive set addition — preserved slots show purple "From previous run" badge. Conflict severity badges. Clearance memo generation.	Complete
Legal Calendar <i>/calendar</i>	Full month-grid with animated hover. Event types: Hearing (blue), Deadline/Drop-dead (red), General (grey). Portal-	Complete

	rendered tooltips avoiding z-index clipping. Add/Delete events. CNR sync auto-populates hearing dates from eCourts.	
Case Vault <i>/vault</i>	Infinite recursive folder tree (create, rename, move, delete) with circular-parentage guard. Breadcrumb navigation. Document grid with 3-dots context menu (Analyze, Summarize, Extract Facts, Download, Delete). FolderMoveModal. Audit trail per document.	Complete
Document Viewer <i>/vault/document/:id</i>	Split-panel — left: paper-styled document with contentEditable body and light-theme markdown. Right: AI analysis sidebar. Rich-text toolbar (Bold, Italic, Underline, Strikethrough, Lists). Floating animated Save FAB. isDirty guard with beforeunload warning and navigation confirmation.	Complete
Virtual Courtroom <i>/war-room</i>	5-stage animated pipeline loader. Live Indian Kanoon precedent search. AI-drafted strategic opening argument. Red-team opposing counsel simulation. Results: Issues cards, Arguments panel, Rebuttal cards, Precedents rail. Export to Vault. Triggered from direct route or via LLM intent from Universal Agent.	Complete

4. Feature Completeness

Status: ✓ Done ⚠ Partial / Known Issue ○ Not Yet Built

4.1 Authentication & Access

- ✓ JWT login with localStorage persistence
- ✓ Sign-up page with form validation
- ✓ Secure sign-out (clears all tokens)
- ⚠ Email verification on sign-up
- Forgot password / password-reset flow
- Google OAuth / Bar Council SSO integration
- Multi-user firm accounts with role-based access

4.2 Universal Agent (AI Chat)

- ✓ Server-Sent Events (SSE) real-time streaming responses
- ✓ Multi-session history — pin, rename, share, delete
- ✓ File upload mid-conversation (PDF, DOCX, images)
- ✓ Draft document generation with rich-text editor toolbar
- ✓ Save-to-Vault from chat (Native / PDF / DOCX format selector)
- ✓ AI provenance audit trail (agent session linked to saved doc)
- ✓ LLM intent routing — navigates to any module from chat
- ✓ Courtroom simulation trigger from Universal Agent

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- Voice input (microphone button)
 - Inline citation hyperlinks to Indian Kanoon judgments

4.3 Case Vault

- ✓ Infinite recursive folder nesting (adjacency-list model)
- ✓ Rename, Move, Delete folders with circular-parentage guard
- ✓ Document 3-dots context menu (Analyze, Summarize, Extract, Download, Delete)
- ✓ contentEditable inline document editing with dirty-state guard
- ✓ Floating Save FAB with animation and unsaved-changes warning
- ✓ PDF and DOCX export on save (stored as BLOB in SQLite)
- ✓ Explorer-style drill-down Save Modal with breadcrumb navigation
- ✓ Folder badge shown on document cards
- Full-text search across all vault documents
- Bulk select, bulk move, bulk delete, bulk download as ZIP
- Download binary file (PDF/DOCX) from vault card menu

4.4 Contract Analyzer

- ✓ PDF and DOCX upload with server-side text extraction
- ✓ AI risk scoring with colour-coded High / Amber / Green pills
- ✓ Clause-level annotation sidebar with inline highlighting
- ✓ AI-powered clause rewrite with one-click replace
- ✓ Collapsible summary banner and Recommendations tab
- ✓ Chat-with-contract panel for ad-hoc questions
- ⚠ Export does not retain original document formatting
- Redline / diff view comparing original vs rewritten clause
- Template library for standard Indian law clause variants

4.5 Conflict Engine

- ✓ Entity name conflict check against case database
- ✓ Batch multi-file upload (drag-and-drop + file picker)
- ✓ Deduplication guard with yellow warning toast
- ✓ Progressive set addition — preserved slots with purple badge
- ✓ Conflict severity: Direct Conflict / Potential / Monitor / Clear
- ✓ Breadcrumb chain display per conflict result
- ✓ Clearance memo auto-generation and save to Vault
- Historical conflict check log / search archive
- Client intake form triggered from conflict-clear result

4.6 Virtual Courtroom

- ✓ 5-stage animated pipeline loader with per-stage status dots
- ✓ Live Indian Kanoon precedent search integrated in pipeline
- ✓ Strategic opening argument drafted by AI
- ✓ Opposing-counsel AI red-team with rebuttal cards
- ✓ Results page: Issues, Arguments, Rebuttals, Precedents tabs
- ✓ Export full simulation package to Case Vault
- Interactive turn-by-turn debate mode (lawyer types, AI rebuts)
- Judge AI scoring and ruling simulation

4.7 Legal Calendar

- ✓ Full month-grid calendar with event dot indicators
- ✓ Portal-rendered hover tooltip (avoids z-index / overflow clipping)
- ✓ Add event modal with type, date, title, case-link fields
- ✓ Delete events from day view
- ✓ CNR sync auto-populates hearing dates from eCourts API
- ✓ Drop-dead / limitation deadline shown in red accent
- Recurring event support (weekly hearings)
- Email and in-app push reminders before deadlines
- Google Calendar / Outlook two-way sync

4.8 Shell & UX System

- ✓ Persistent sidebar with 7 primary nav items + live case list
- ✓ Topbar breadcrumbs auto-generated from current route
- ✓ Light / Dark / Light-Matte theme toggle, persisted to localStorage
- ✓ Focus Mode — CSS-only sidebar hide with smooth 300ms transition
- ✓ Focus Mode escape hatch button (fixed position, outside sidebar)
- ✓ Ctrl+ / Cmd+ global keyboard shortcut to toggle Focus Mode
- ✓ Mobile hamburger + off-canvas drawer with overlay
- ✓ Page-enter animation (fade + slide) on every route change
- Global toast / notification system for async feedback
- Onboarding / empty-state coaching for new users

5. Gaps & Prioritised Build List

The following 12 items represent everything that has not yet been built, ranked by deployment impact. P1 items block real-world usage by lawyers. P2 items block firm-level adoption. P3 items are polish.

#	Module	Gap Description	Priority
01	Case Vault	Download button for PDF/DOCX blobs — backend route /api/vault/documents/<id>/download exists but is not yet wired to the VaultView card context menu.	P1 Critical
02	Auth	Password reset / forgot-password flow is completely absent. Any lawyer who forgets their password has no recovery path — a hard deployment blocker for real users.	P1 Critical
03	Case Vault	Full-text search across all vault documents. Currently only folder navigation exists. Lawyers need to find documents by keyword, case name, or tag across the entire vault.	P1 Critical
04	Calendar	No notification or reminder system. Events are stored but nothing alerts the lawyer before a hearing or deadline. Email / in-app push reminders are essential for any legal deadline tool.	P1 Critical
05	Contract Analyzer	Redline / diff view when a clause is rewritten by AI. Currently the rewrite replaces text with no visual comparison to the original. Lawyers need before/after comparison or tracked-changes view.	P2 High
06	Auth	Multi-user / firm-level accounts. All data is currently per-device with no concept of a firm, team, or shared vault. Required before any firm-level sale.	P2 High
07	Virtual Courtroom	Interactive turn-by-turn debate mode — the lawyer types a response, the opposing AI rebuts. Currently the simulation is a single one-shot generation, not a real conversation.	P2 High
08	Case Vault	Bulk select operations — checkboxes, select-all, bulk move-to-folder, bulk delete, bulk download as ZIP. Standard file-manager behaviour lawyers expect from a Windows-explorer-style UI.	P2 High
09	Universal Agent	Inline citation hyperlinks — when the LLM cites a judgment, the text should be a clickable link to Indian Kanoon rather than plain text. Builds verifiability and trust.	P2 High
10	Calendar	Google Calendar / Outlook two-way sync. Lawyers use external calendars on mobile. Hearing dates added via CNR	P3 Medium

		sync should propagate automatically to their existing calendar.	
11	Shell	Onboarding flow for new users — empty-state screens with guided first-action prompts (e.g. "Upload your first contract", "Sync your first CNR case"). Currently new users land on a blank dashboard.	P3 Medium
12	Conflict Engine	Search history / log for past conflict checks. Lawyers need to go back to a previous clearance memo without re-running the full check. No persistence of check history currently.	P3 Medium

7. Recommended Next Steps

7.1 Immediate (P1 — Ship Blockers)

- Wire the download endpoint — connect `/api/vault/documents/<id>/download` to the VaultView card menu for PDF and DOCX files.
- Build password reset flow — email-based token reset is the minimum; without it no lawyer can recover their account.
- Add vault full-text search — a search bar above the document grid that queries content, `smart_title`, and tags columns via SQLite FTS5.
- Implement reminder notifications — at minimum, a daily digest email of deadlines in the next 7 days, using `calendar_events` table data.

7.2 Short-Term (P2 — Firm Adoption)

- Contract redline diff view — render a before/after split or track-changes-style highlight when a clause is rewritten.
- Multi-user accounts — introduce a firms table, user-to-firm membership, and shared vault with per-user write guards.
- Interactive courtroom debate — convert the one-shot simulation into a turn-by-turn streaming conversation with the AI opponent.
- Bulk vault operations — checkboxes on document cards, select-all header, bulk move/delete/download as ZIP.
- Citation hyperlinks in AI — post-process LLM output to detect citation patterns and wrap them in Indian Kanoon deep-links.

7.3 Medium-Term (P3 — Polish)

- Google Calendar / Outlook sync via OAuth — two-way so hearing dates appear in lawyers' existing mobile calendars.
- Onboarding flow — guided empty states for first-time users with step-by-step prompts for first contract upload, first CNR sync, first vault save.

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- Conflict check history log — persist each check with its entity names, documents ingested, results, and clearance memo link.
 - Unified global toast system — replace per-component toast state with a single context-driven toast queue for consistent in-app feedback.
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